DARK WEB THREAT INTELLIGENCE SCRAPER AND AUTOMATED THREAT CLASSIFY USING ML

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AD22811 – PROJECT WORK

First Review

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PROBLEM STATEMENT

- Cyberattacks are often preceded by early indicators such as vulnerability discussions, exploit sharing, malware advertisements, and credential leaks across underground and threat-intelligence sources. Existing security solutions primarily operate in a reactive manner, detecting attacks only after network intrusion or data compromise has occurred. This results in delayed response and increased damage.
- The challenge is to design and develop a proactive cyber threat detection system that can collect threat-related data from multiple sources, process large volumes of unstructured text, and apply machine learning techniques for accurate threat classification and risk assessment. The system should leverage computer networking principles for secure data transmission and real-time alert generation. The proposed solution must prioritize threats based on severity and provide timely notifications to support early decision-making and preventive cybersecurity measures.

INTRODUCTION

Early Detection of Cyberattacks Using Dark Web Intelligence and Intelligent Alerting

With the increasing dependency on digital systems and online services, cyberattacks have become more advanced, coordinated, and difficult to detect. A significant portion of these attacks is planned and discussed on the Dark Web through underground forums, marketplaces, and leak platforms. Conventional cybersecurity solutions mainly operate in a reactive manner, identifying threats only after system compromise, which leads to data loss, service disruption, and financial damage.This project proposes an early cyberattack detection system using Dark Web intelligence combined with machine learning–based threat analysis and IoT-enabled alerting.

ISSUES AND CHALLENGES

- **Early Identification of Ransomware Threats**
  Ransomware campaigns are often discussed in coded language on Dark Web forums, making early detection and accurate interpretation difficult.

Dynamic and Hidden Dark Web Sources
Ransomware groups frequently change their communication channels and leak sites, which complicates continuous intelligence gathering.

- **Encrypted and Obfuscated Content**
  Ransomware-related discussions may use encryption, aliases, or obfuscated keywords to evade detection, reducing text analysis accuracy.
- **Limited Availability of Ransomware Training Data**
  High-quality, labeled datasets specific to ransomware threats are scarce, affecting machine learning model effectiveness.
- **False Positives in Ransomware Detection**
  Discussions about ransomware research or defensive analysis can be misclassified as real threats, increasing false alerts.

ISSUES AND CHALLENGES

•**Real-Time Alerting Challenges**
Ransomware attacks require rapid response; delays in detection, analysis, or IoTalert triggering reduce system usefulness.

•**IoT Alert Reliability During High-Risk Events**
Network interruptions during ransomware incidents may impact communication between the threat server and IoT devices.

•**Secure Transmission of Threat Intelligence**
Ransomware-related intelligence must be transmitted securely to prevent interception or manipulation by attackers.

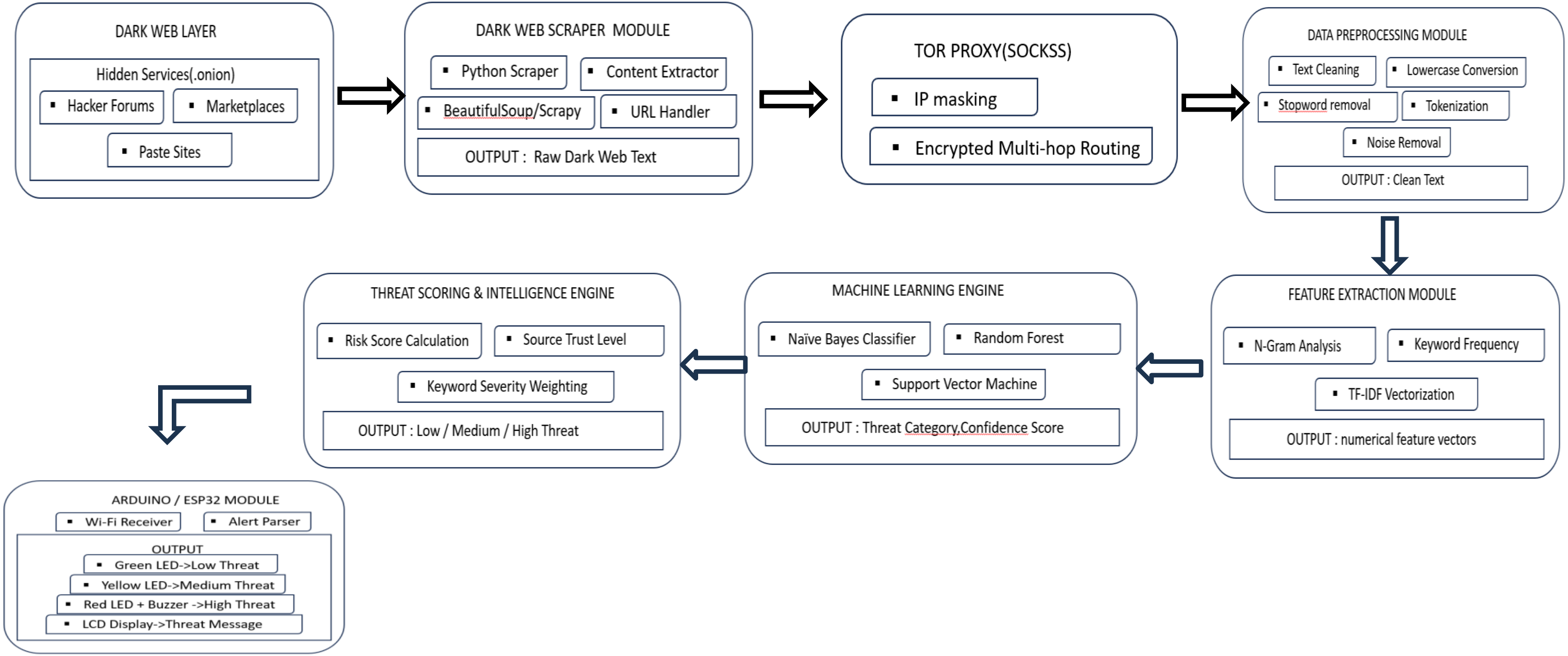
•**Hardware Constraints of IoT Devices**
Limited processing power of Arduino-based systems restrict on-device threat analysis,relying heavily on backend servers.

•**Operational and Legal Considerations**
Handling ransomware intelligence requires strict compliance with cybersecurity laws and ethical standards.

LITERATURE REVIEW

- The increasing complexity of cyber threats has driven researchers to explore the dark web as a critical source of threat intelligence. The dark web hosts forums, marketplaces, and communication channels where cybercriminals exchange malware, stolen credentials, exploit kits, and attack strategies. Existing literature emphasizes the use of specialized dark web crawlers capable of operating within anonymized networks such as TOR. Due to the unstructured and noisy nature of dark web data, researchers propose robust data preprocessing techniques, including data cleaning, normalization, and anonymization. Natural Language Processing (NLP) methods are widely applied to extract meaningful entities and topics from forum discussions, while machine learning models assist in classifying malicious content and identifying emerging threat patterns. Several studies also focus on correlating dark web intelligence with organizational security logs to enhance threat accuracy. Integration with SIEM and SOAR platforms is identified as a key factor in operationalizing intelligence and enabling automated alerting and response.

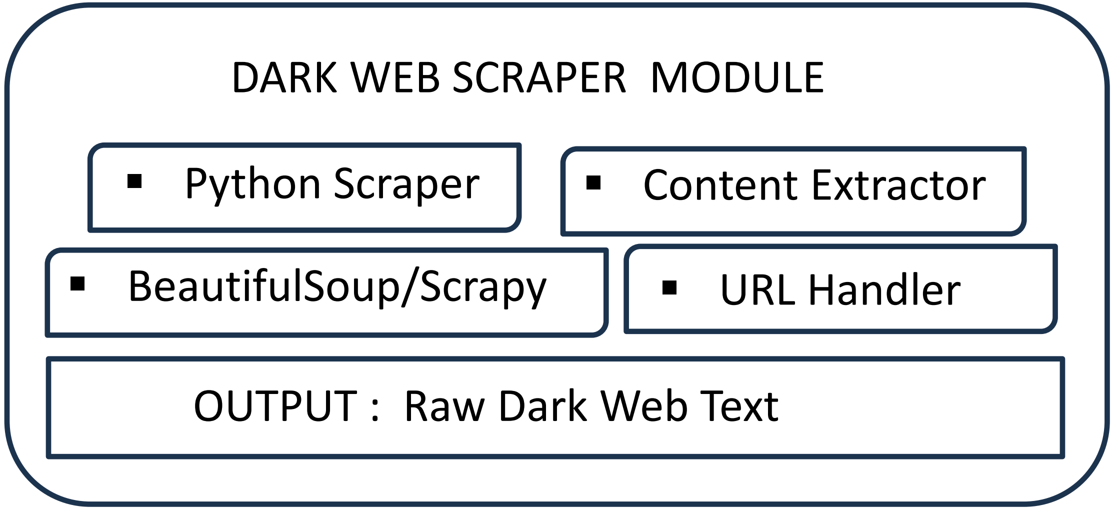
PROPOSED WORK-ARCHITECTURE DIAGRAM



SYSTEM REQUIREMENTS AND TOOLS

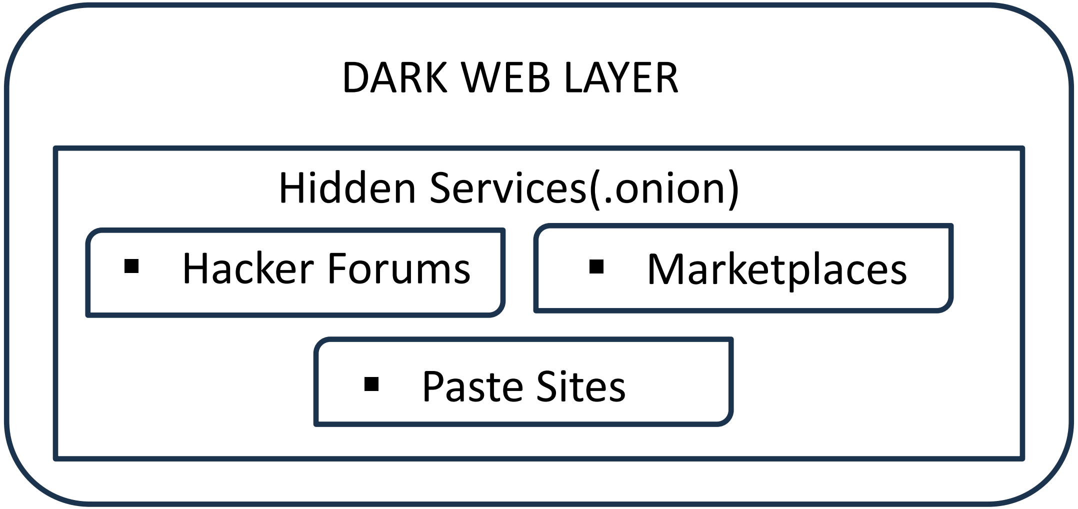
- The Dark Web Threat Intelligence System requires a multi-core CPU (Intel i7/AMD Ryzen 7), 16–32 GB RAM, high-speed SSD (≥512 GB), stable internet, and optional GPU for ML/NLP tasks. Software includes Python 3.9+, Linux/Windows, and optionally Java/Node.js. Databases like MongoDB/MySQL/PostgreSQL store structured data; visualization via Power BI, Matplotlib, or Plotly. Tor Browser/Proxy enables secure dark web access. Scrapy, BeautifulSoup, Selenium, and Requests handle crawling; NLP/ML with NLTK, SpaCy, Gensim, Scikit-learn, and Transformers; topic modeling via BERTopic/LDA.

MODULES AND DESCRIPTION



The Dark Web Scraper Module collects data from hidden web sources using Python-based scrapers. It extracts content, manages URLs, and leverages tools like BeautifulSoup and Scrapy. The module outputs raw dark web text for further processing and analysis.

MODULES AND DESCRIPTION



The Dark Web Scraper Module is responsible for collecting hidden web data using Python-based scrapers. It utilizes tools like BeautifulSoup and Scrapy for content extraction and handles URLs efficiently. The module outputs structured raw dark web text for further analysis in the intelligence pipeline.

MODULES AND DESCRIPTION

MACHINE LEARNING ENGINE

- Naïve Bayes Classifier
- Random Forest
- Support Vector Machine

OUTPUT : Threat Category,Confidence Score

Naïve Bayes Classifier: A probabilistic model that predicts threat categories based on feature likelihoods, assuming feature independence.

Random Forest: An ensemble learning method using multiple decision trees to improve classification accuracy and reduce overfitting.

MODULES AND DESCRIPTION

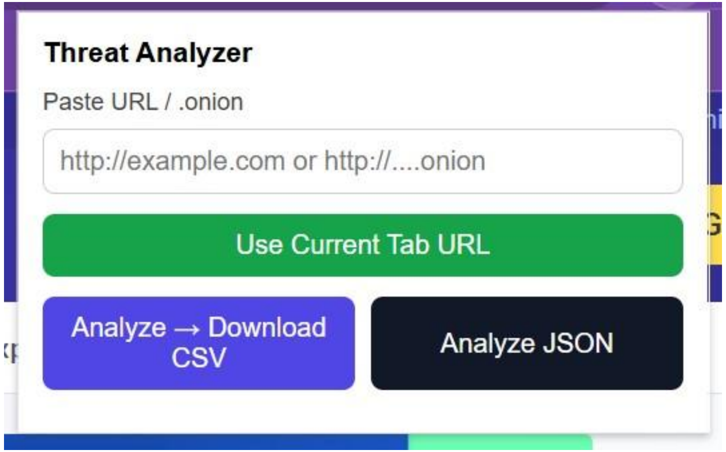
ARDUINO / ESP32 MODULE

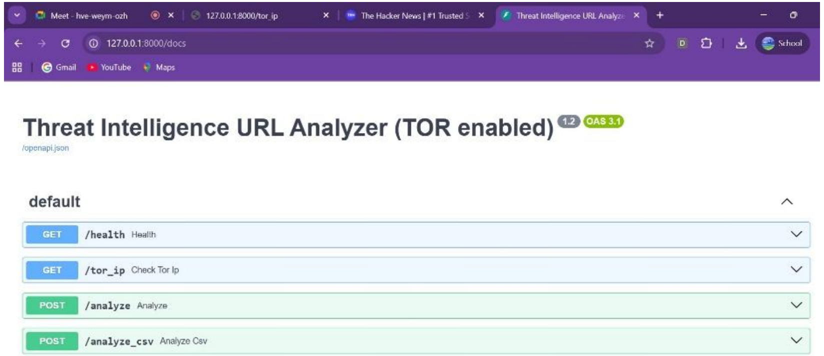
- Wi-Fi Receiver
- Alert Parser

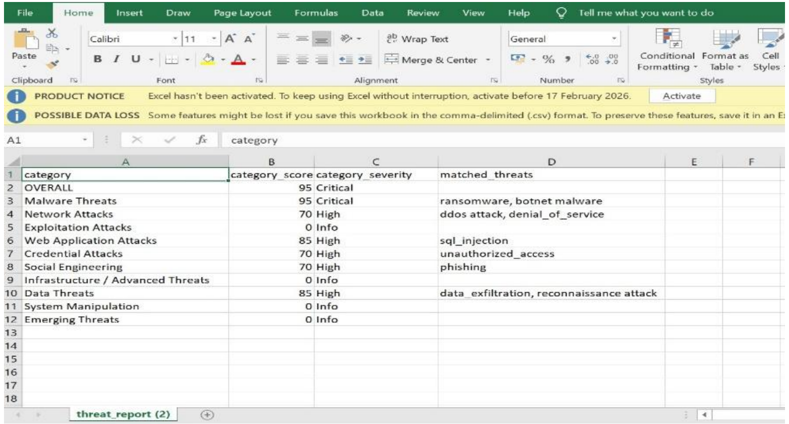
OUTPUT

- Green LED->Low Threat
- Yellow LED->Medium Threat
- Red LED + Buzzer ->High Threat
- LCD Display->Threat Message

IMPLEMENTATION







category	category_score	category_severity	matched_threats
OVERALL	95	Critical	
Malware Threats	95	Critical	ransomware, botnet malware
Network Attacks	70	High	ddos attack, denial_of_service
Exploitation Attacks	0	Info	
Web Application Attacks	85	High	sql_injection
Credential Attacks	70	High	unauthorized_access
Social Engineering	70	High	phishing
Infrastructure / Advanced Threats	0	Info	
Data Threats	85	High	data_exfiltration, reconnaissance attack
System Manipulation	0	Info	
Emerging Threats	0	Info	

REFERENCES

- **[1] Base Paper**

Yazdanjue, N., et al.,“Cyber threat management using semi-supervised ensemble learning and enhanced interior search algorithm: applications for illicit marketplace classification in deep/dark web and social platforms,”Annals of Operations Research,Springer Nature, 2025.

- **[2] Supporting Reference**

Mouiche, I., and Saad, S.,“Entity and relation extractions for threat intelligence knowledge graphs,”Computers & Security,Elsevier, 2025.

- **[3] Supporting Reference**

Samtani, S., et al.,“Leveraging machine learning for cyber threat intelligence extraction,”ACM Transactions on Privacy and Security,Association for Computing Machinery (ACM), 2025.